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FRUGAL EVALUATION OF RECOMMENDER SYSTEMS IN TERRITORIAL
MARKETING

ENSAIO DE TESE

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Frugal Evaluation of Recommender Systems in Territorial Marketing

Abstract. This theoretical essay problematizes the assumption that superior algorithmic performance necessarily generates greater marketing value when recommender systems support territorially situated decisions. It proposes frugality as an evaluative lens for assessing these systems beyond isolated performance metrics such as accuracy, ranking quality, robustness, and scalability. By connecting recommender systems and territorial marketing through this lens, drawing on selected concepts from Green AI, Frugal AI, and frugal innovation, the essay develops a relational framework organized around three interdependent dimensions: functional adequacy, resource efficiency, and strategic coherence. Functional adequacy concerns the extent to which the system's capabilities and outputs correspond to the spatial, temporal, contextual, and decisional conditions of the territorial problem. Resource efficiency refers to the proportionality between the value produced and the computational, energy, financial, environmental, and operational resources mobilized throughout the system lifecycle. Strategic coherence concerns the alignment between recommendations and the organizational, institutional, and territorial objectives that motivate system adoption. The model associates these dimensions with decision usefulness, implementation viability, and decision-support quality, respectively, and their joint, interdependent operation constitutes the frugal value of the territorial recommender system. The essay advances a relating contribution, specifically within the integrating subgoal identified by MacInnis (2011), by revealing an evaluative structure that remains obscured when these concepts are examined in isolation. It demonstrates that technical performance is translated into marketing value only when it is functionally adequate, materially justifiable, and strategically coherent. The theoretical development was supported by a structured literature mapping documented in an open-science repository.

Keywords: recommender systems; territorial marketing; frugality; Green AI; Frugal AI; decision support.

Resumo. Este ensaio teórico problematiza a premissa de que um desempenho algorítmico superior gera necessariamente maior valor de marketing quando sistemas de recomendação apoiam decisões territorialmente situadas. Propõe a frugalidade como uma lente avaliativa para examinar esses sistemas para além de métricas isoladas de desempenho, como acurácia, qualidade do ranking, robustez e escalabilidade. Ao conectar sistemas de recomendação e marketing territorial por meio dessa lente, mobilizando conceitos selecionados da Green AI, da IA Frugal e da inovação frugal, o ensaio desenvolve um framework relacional organizado em três dimensões interdependentes: adequação funcional, eficiência de recursos e coerência estratégica. A adequação funcional diz respeito ao grau em que as capacidades e os resultados do sistema correspondem às condições espaciais, temporais, contextuais e decisórias do problema territorial. A eficiência de recursos refere-se à proporção entre o valor produzido e os recursos computacionais, energéticos, financeiros, ambientais e operacionais mobilizados ao longo do ciclo de vida do sistema. A coerência estratégica refere-se ao alinhamento entre as recomendações e os objetivos organizacionais, institucionais e territoriais que motivam a adoção do sistema. O modelo associa essas dimensões, respectivamente, à utilidade decisória, à viabilidade de implementação e à qualidade do apoio à decisão, e sua operação conjunta e interdependente constitui o valor frugal do sistema de recomendação territorial. O ensaio apresenta uma contribuição do tipo relating, especificamente no subobjetivo de integrating identificado por MacInnis (2011), ao revelar uma estrutura avaliativa que permanece obscurecida quando esses conceitos são examinados isoladamente. Demonstra-se que o desempenho técnico somente se converte em valor de marketing quando é funcionalmente adequado, materialmente justificável e estrategicamente coerente. O desenvolvimento teórico foi apoiado por um mapeamento estruturado da literatura, documentado em um repositório de ciência aberta.

Palavras-chave: sistemas de recomendação; marketing territorial; frugalidade; Green AI; IA Frugal; apoio à decisão.

1. INTRODUCTION

Artificial intelligence (AI) has become increasingly prominent in marketing strategies, particularly in data-intensive digital environments. However, this development is neither immaterial nor neutral. It depends on increasingly energy-intensive computational infrastructures, especially data centers, whose environmental and operational costs often remain relatively invisible in marketing discussions. In North America, for example, data-center power demand increased from 2,688 megawatts at the end of 2022 to 5,341 megawatts in 2023 and could reach 1,050 terawatt-hours by 2026, an amount comparable to the annual electricity consumption of approximately twenty cities the size of New York. This expansion has been intensified by the rapid diffusion of AI, particularly large-scale generative models, which require substantial computing power, high electricity consumption, and large volumes of water for cooling, and generate carbon dioxide emissions that contribute to broader environmental pressures (Zewe, 2025). These concerns are consistent with recent energy-sector assessments indicating that data centers consumed approximately 415 TWh in 2024, equivalent to about 1.5% of global electricity consumption, and that this demand may reach approximately 945 TWh by 2030 (International Energy Agency, 2025). They are also reinforced by warnings that data centers may place increasing pressure on water and energy systems, especially as medium- and large-scale facilities may reach electricity consumption levels comparable to those of thousands of households (Borges, 2025).

This material dimension of AI is particularly relevant to marketing because algorithmic systems increasingly support segmentation, personalization, performance measurement, automation, and decision-making. In data-rich environments, marketing analytics has become a central infrastructure through which organizations transform large volumes of information into more precise and actionable decisions (Wedel; Kannan, 2016). Davenport et al. (2020) argue that AI may reshape marketing by enhancing analytical capabilities and automating processes and decisions, with implications for costs, revenues, and organizational efficiency. Kumar *et al.* (2024) similarly emphasize the role of AI in personalization, value creation, and marketing decision support. Within this broader transformation, recommender systems occupy a particularly important position. They may be understood as algorithmic systems designed to support choice by filtering large volumes of information and suggesting or ranking relevant alternatives

according to users, items, preferences, behaviors, or contextual attributes (Ricci *et al.*, 2011).

In marketing, recommender systems are not merely technical tools for personalization. They operate as decision-support infrastructures that influence what becomes visible, relevant, and actionable in contexts characterized by information overload and choice complexity. By ranking products, services, content, places, or opportunities, these systems help construct market relevance and prioritize alternatives. Their value, however, is commonly assessed through technical performance indicators such as accuracy, predictive quality, ranking precision, scalability, and computational sophistication. Although these criteria are important, they may be insufficient when used as the primary basis for evaluating the contribution of recommender systems to marketing decisions.

This limitation becomes clearer when the performance-oriented logic of recommender systems is examined in light of the Green AI debate. Schwartz *et al.* (2020) argue that artificial intelligence research has often prioritized improved results through greater computational power, larger datasets, and more intensive experimentation. Strubell *et al.* (2019) similarly demonstrate that the development and use of advanced models may involve substantial financial, energy, and environmental costs. These contributions suggest that the quality of an AI-based system should be assessed not only by whether it performs better, but also by the resources required to achieve that improvement. For marketing, this is a central issue: a technically superior system is not necessarily the most appropriate solution for a specific decision problem when its gains are marginal, costly, difficult to interpret, or poorly aligned with the strategic purpose of its use.

This tension becomes even more relevant when recommender systems are applied to territorially situated decisions. In this essay, territorial marketing is understood as a strategic marketing perspective concerned with the interpretation, organization, positioning, and activation of places, regions, and territories as objects of attractiveness and value creation. Kavaratzis (2005) demonstrates that place-related marketing extends beyond promotion and involves the development of coherent territorial value propositions for different audiences, including residents, visitors, investors, organizations, and public and private actors. Warnaby (2009) and Warnaby and Medway (2013) further emphasize that places are complex, relational, and multifaceted entities whose value depends on resources, actors, structures, meanings, narratives, and experiences. Therefore, when

recommendations concern territories, the problem is not simply one of suggesting a product, service, or piece of content. Rather, it involves supporting decisions regarding spatial alternatives, territorial attractiveness, opportunity identification, market prioritization, expansion, and activation.

For this reason, recommender systems in territorial marketing cannot be adequately evaluated solely on algorithmic performance. A recommendation may be technically accurate yet have limited marketing value if it fails to address the territorial problem it is intended to support. For example, a system may rank locations efficiently while ignoring spatial constraints, contextual meanings, market objectives, or the strategic relevance of the recommended alternatives. Conversely, a less complex system may generate greater value by providing sufficient, intelligible, and contextually appropriate support for territorial decisions while using fewer computational, energy, and operational resources. The central issue, therefore, is not whether performance matters, but whether performance alone is sufficient to define the value of recommender systems in territorial marketing.

This essay argues that it is not. Its central thesis is that, in territorial marketing contexts, the value of recommender systems should not be equated with isolated algorithmic performance. Instead, these systems should be evaluated according to their capacity to generate territorially meaningful decision support across three interconnected dimensions: functional adequacy, resource efficiency, and strategic coherence. Functional adequacy refers to the extent to which the system responds appropriately to the territorial problem it is intended to address. Resource efficiency refers to the justifiable use of computational, energy, and operational resources relative to the value produced by the recommendation. Strategic coherence refers to the alignment between the system's recommendations and the broader objectives of territorial marketing, such as prioritization, expansion, territorial activation, and spatial intelligibility.

Frugality provides the theoretical lens through which these dimensions can be integrated. In the frugal innovation literature, value is not associated with the unrestricted maximization of sophistication, features, or performance, but with the capacity to respond to relevant needs under resource constraints while preserving core functionalities and sufficient quality (Bhatti; Ventresca, 2013; Weyrauch; Herstatt, 2016). In this sense, frugality does not imply technological inferiority or a rejection of AI. Rather, it introduces a normative evaluative principle: a superior recommender system is not necessarily the most complex or computationally intensive one, but the one that provides adequate,

efficient, and strategically meaningful support for the decision problem at hand. Applied to territorial marketing, this means asking not only whether a system recommends accurately, but also whether it recommends territorially relevant alternatives using justifiable resources and in alignment with the objectives that motivated its adoption.

The research gap addressed in this essay stems from the fact that these debates have largely developed in parallel. The AI and marketing analytics literature has emphasized personalization, automation, and decision support (Davenport *et al.*, 2020; Kumar; *et al.*, 2024; Wedel; Kannan, 2016). The recommender systems literature has examined how systems filter, rank, and suggest alternatives under conditions of information overload (Ricci *et al.*, 2011). The Green AI and frugality literatures have questioned the cost, efficiency, and sufficiency of technological solutions (Bhatti; Ventresca, 2013; Schwartz *et al.*, 2020; Strubell *et al.* (2019); Weyrauch; Herstatt, 2016). The territorial marketing literature, in turn, has shown that places are strategic, relational, and complex objects of value creation (Kavaratzis, 2005; Warnaby, 2009; Warnaby; Medway, 2013). What remains underdeveloped is a conceptual explanation of how recommender systems should be evaluated when used to support territorial marketing decisions, where their value depends not only on predictive performance but also on territorial relevance, resource use, and strategic alignment.

Accordingly, the guiding question of this essay is: how can the evaluation of recommender systems in territorial marketing be reoriented through a frugal lens? To address this question, the essay develops a conceptual framework for the frugal evaluation of recommender systems in territorial marketing. The framework is organized around three dimensions: functional adequacy, resource efficiency, and strategic coherence. These dimensions are not proposed as a simple list of desirable attributes, but as interdependent evaluative conditions that challenge the assumption that higher algorithmic performance automatically produces greater marketing value in territorial contexts.

The contribution of this essay is conceptual. Drawing on MacInnis's (2011) typology of theoretical contributions, the essay develops a relating contribution, more specifically an integrating contribution, by connecting recommender-system evaluation, frugality, and territorial marketing within a single analytical structure. Rather than merely extending the existing literature to an underexplored empirical context, the essay problematizes a taken-for-granted assumption: that algorithmic performance can be treated as equivalent to marketing value. In line with Alvesson and Sandberg's (2011)

argument on problematization, the objective is not only to identify a gap, but also to question the evaluative logic through which recommender systems are commonly assessed.

The theoretical development is informed by a structured mapping of the literature, documented in an open-science repository that includes the search protocol, classification procedures, and supporting materials. However, because the essay's purpose is theoretical rather than empirical, the review serves as a basis for conceptual development rather than as the study's main contribution. The remainder of the essay is organized as follows. First, it discusses recommender systems and their role in supporting marketing decisions. It then examines territorially situated recommendation problems and clarifies the distinction among generic recommendation, place recommendation, and territorial decision support. Next, territorial marketing is presented as the strategic context in which recommendations acquire territorial meaning. The essay then develops the frugal evaluation framework and its three dimensions. Finally, it presents the conceptual model and the research propositions derived from this theoretical integration.

2. METHOD

The theoretical development of this essay was supported by a structured mapping of the literature, used as a basis for conceptual construction rather than as the study's primary contribution. This orientation is consistent with an integrative review approach aimed at combining perspectives from different research traditions to develop a new conceptual framework (Snyder, 2019). The search, screening, and organization of the studies were conducted systematically using Web of Science and Scopus, supported by computational procedures implemented in R. The full search protocol, Boolean search strings, eligibility criteria, selection flow, document corpus, scripts, and traceability materials are available in a public open-science repository. This documentation follows recommendations that literature review procedures and the rationales underlying methodological decisions be reported transparently and reproducibly (Paul *et al.*, 2021). Within the essay, the literature is mobilized interpretively and integratively to connect theoretical arguments with the evidence synthesized, rather than merely cataloging the studies identified in the mapping (Siddaway, Wood, e Hedges, 2019). This synthesis is used to problematize the equivalence between algorithmic performance and marketing value and to derive the conceptual relationships that structure the proposed framework.

This approach preserves the transparency of the literature-mapping process without turning the qualification essay into a systematic-review article.

2.1 Artificial Intelligence, Recommender Systems, and Performance-Oriented Evaluation in Marketing

Artificial intelligence has expanded the analytical and operational capabilities of marketing by supporting segmentation, personalization, forecasting, automation, and decision-making in data-intensive environments. Marketing analytics transforms large and heterogeneous datasets into information that guides organizational action, while AI extends this capacity through systems that learn patterns, update predictions, and automate parts of the decision-making process (Wedel; Kannan, 2016; Davenport *et al.*, 2020; Kumar *et al.*, 2024). In this process, recommender systems play a particularly important role, as they filter, rank, and suggest alternatives based on information about users, items, behaviors, preferences, and context (Ricci *et al.*, 2011).

In marketing, recommender systems extend beyond the personalization of offerings. By influencing which products, services, content, places, or opportunities become visible and relevant, they operate as decision-support infrastructures. Their evaluation therefore commonly focuses on metrics such as accuracy, precision, recall, ranking quality, robustness, scalability, and response time. These indicators remain indispensable because they demonstrate whether the system performs its technical task effectively. The theoretical problem arises when technical performance is treated as a sufficient proxy for marketing value, without considering whether the recommendation is appropriate for the decision problem, whether the required resources are justifiable, and whether the resulting outputs support the strategic purpose that motivated adoption.

The Green AI debate makes the material foundations of this evaluative logic visible. Strubell *et al.* (2019) show that the development and fine-tuning of advanced models may involve substantial financial, energy, and environmental costs. Schwartz *et al.* (2020) use the term Red AI to describe an orientation that seeks incremental performance gains through computational intensification while devoting limited attention to efficiency. This argument should not be interpreted as a rejection of high-performing models or as an exhaustive description of the AI field in 2026. Rather, it identifies a persistent evaluative tension that has subsequently been broadened by Frugal AI frameworks, which shift the analysis from the cost of an isolated model to total cost of

ownership, lifecycle efficiency, organizational returns, and portfolio-level impact (Ahluwalia *et al.*, 2025; Gandikota *et al.*, 2025).

Frugal innovation offers a complementary evaluative logic. Rather than equating superiority with the unrestricted maximization of features or performance, it emphasizes substantial cost reduction, a focus on core functionalities, and performance optimized for the context of use (Bhatti; Ventresca, 2013; Weyrauch; Herstatt, 2016). This does not mean accepting insufficient performance. It means assessing whether additional complexity generates meaningful value for the decision or merely increases the resources required to obtain marginal gains. Albert (2022) also cautions that frugality should not be automatically equated with environmental sustainability, as ecological effects depend on the solution's configuration and use.

Taken together, these arguments support a shift from performance-oriented evaluation to relational evaluation. Accuracy and other technical metrics remain necessary, but they must be interpreted in relation to three additional questions: Is the system functionally adequate to the problem? Are the resources consumed proportional to the value produced? Is the recommendation strategically coherent with the objectives underlying its use? The territorial context developed in the following sections makes this tension especially evident, as the recommended alternatives are spatially situated and strategically consequential rather than merely interchangeable digital items.

2.2 Place Recommendation as a Territorially Situated Problem

When recommendation involves places, the problem is no longer limited to an abstract match between user and item. In broad terms, place recommendation may be understood as the selection and ranking of spatially situated alternatives based on information about users, locations, and the context of use. Within this domain, point-of-interest (POI) recommendation constitutes a specific application, typically aimed at suggesting places that an individual user may wish to visit. In this literature, a POI is a location identified by geographic and semantic attributes, and the task consists of recommending places of interest by considering the user's current location, visit history, time, and preferences (Yin *et al.*, 2015; Cai *et al.*, 2018).

The specificity of POI recommendation stems from the fact that human behavior in physical space is constrained in ways that do not apply with the same intensity to recommendations of movies, books, or digital content. Accessibility, distance, travel sequence, and time of visit affect both the feasibility and relevance of an alternative. Cai

et al. (2018) show that the joint incorporation of spatial and temporal contexts improves recommendation performance, with the spatial component exerting a particularly strong influence in the datasets analyzed. The relevance of a place, therefore, depends not only on the abstract affinity between the user and the alternative, but also on whether that alternative is realistically accessible in the context in which the decision is made.

Yin *et al.* (2015) deepen this understanding by jointly modeling interests, semantic patterns, temporality, and mobility. The problem they address is the high sparsity of the user–POI matrix, especially for users outside their home cities. Their proposed solution demonstrates that individual interests and mobility patterns must be considered together, because places with similar content may assume different levels of relevance depending on the region, time of day, and the user’s spatial history. In this context, the term *item* refers specifically to the POI, that is, the place that may be recommended, rather than to a generic product detached from space.

Baral and Li (2017) likewise demonstrate that POI recommendation depends on the combination of multiple factors. Geographic coordinates, distance, check-in time, place category, visit frequency, ratings, and social relationships may all influence place selection. The authors show that these factors are neither necessarily equivalent nor independently sufficient. Categorical influence, for example, may complement recommendation, but it does not replace the need to incorporate spatial, temporal, and behavioral components. This literature therefore supports the conclusion that the quality of a territorially situated recommendation depends on the system’s sensitivity to the specific configuration of the problem.

Subsequent research reinforces this conclusion by showing that preferences and mobility vary across users, regions, and times. Static models based solely on check-in histories may generate impractical recommendations when they ignore the user’s current location and time. Likewise, similarity measures that disregard proximity, trajectories, preferred regions, and social relationships provide an impoverished representation of the phenomenon. POI recommendation thus emerges as an analytical practice that is simultaneously spatial, temporal, semantic, social, and contextual, rather than as a simple inference within an interaction matrix.

This conclusion, however, does not justify equating consumer-oriented POI recommendation with organizational territorial decision-making. The former is generally directed toward individual convenience, place discovery, and next-visit prediction. The latter involves comparing, selecting, and prioritizing municipalities, regions, areas, or

markets for purposes such as expansion, investment, resource allocation, and territorial activation. The POI literature does not necessarily address these organizational decisions, but it provides relevant foundations concerning spatial, temporal, and contextual sensitivity. In this essay, these foundations are therefore transferred, under an explicit delimitation, from the scale of individual visitation to the scale of decision support in territorial marketing.

The proposed framework is not intended to apply indiscriminately to every form of place recommendation. Its focus is on systems that support organizations and territorial actors in comparing, selecting, prioritizing, expanding, or activating spatially situated alternatives. Applications exclusively directed toward navigation, entertainment, or individual convenience do not constitute the model's primary purpose, although the literature examining them provides important technical principles. The central distinction lies in the purpose of recommendation: in territorial marketing, place is no longer merely the likely destination of an individual user, but becomes a strategic alternative within an organizational decision.

The notion of functional adequacy emerges from the intersection between the literature on territorially situated recommendation and frugal innovation. Frugal innovation directs design toward core functionalities and an optimized level of performance rather than the unrestricted maximization of features (Weyrauch; Herstatt, 2016). In territorial problems, this means the system must preserve the capabilities required to represent space, time, context, and the purpose of the decision. Functional adequacy therefore refers to the extent to which the system's functionalities and outputs respond sufficiently and contextually appropriately to the territorial problem for which it was adopted.

Proposition 1. The decision usefulness of a recommender system applied to territorial marketing increases as its functionalities and outputs become more adequate to the spatial, temporal, contextual, and decisional conditions of the territorial problem under consideration.

2.3 Territorial Marketing as the Strategic Structure of the Problem

Within strategic marketing, decisions concerning location, expansion, market selection, and opportunity identification depend on transforming complex information into actionable intelligence. When such decisions are territorially situated, place cannot be treated merely as a geographic setting, a coordinate, or an object of communication. In this essay, territorial marketing is understood as a strategic practice of interpreting,

organizing, positioning, and activating cities, regions, and territories according to their capacity to attract and retain visitors, residents, organizations, investment, and other forms of flow.

Kavaratzis (2005) shows that the application of marketing to places evolved from fragmented promotional initiatives toward more integrated, strategic, and management-oriented approaches. Competition among cities, regions, and countries for resources, investment, firms, visitors, and residents intensified the need to structure coherent territorial offerings. From this perspective, territory ceases to be a passive object of promotion and requires decisions concerning infrastructure, services, attractiveness, positioning, communication, and institutional coordination.

This shift matters because it prevents territorial marketing from being reduced to advertising. External communication is only one component of a broader process of place management. A territorial proposition must articulate physical attributes, services, uses, and the expectations of distinct audiences. A territory may be visible yet fail to provide conditions consistent with its intended positioning; it may attract attention but fail to sustain the experience, investment, or continued presence of the audiences it seeks. Territorial strategy therefore concerns organizing and activating resources, not merely representing them.

Warnaby (2009) highlights the complexity of both the place product and the organizational mechanisms responsible for its marketing. Unlike a conventional offering controlled by a single organization, places comprise multiple goods, services, institutions, infrastructure, social groups, and experiences. Their management involves public, private, and community actors, often with divergent interests. Consequently, the construction of territorial objectives requires coordination, negotiation, and the formulation of propositions that can articulate heterogeneous resources and expectations.

Warnaby and Medway (2013) deepen this complexity by conceptualizing place as a multifaceted phenomenon composed of material elements and changing, competing narratives. Territory does not possess a single and stable meaning; it is continuously interpreted, represented, and reconstructed by the actors who inhabit, manage, visit, or contest it. This condition problematizes any recommendation that treats place as a homogeneous, neutral alternative fully captured by technical attributes.

A complementary perspective for understanding territorial value is Service-Dominant Logic, developed by Vargo and Lusch (2004, 2008) and specifically mobilized in the place-marketing context by Warnaby (2009). This application does not constitute

a universal premise of the entire territorial-marketing field. It is used in this essay because it clarifies that the value of a place is not fully embedded in its infrastructure or unilaterally delivered by its managers. Value is realized through resource integration, interactions, and the experiences of different audiences with the territory.

Grönroos's (2011) contribution helps make this claim more precise. He criticizes the generic assertion that the customer is always a value co-creator and argues that value is realized as value-in-use by the user. The organization may facilitate this process and, under specific conditions of interaction, participate in co-creation. Carefully applied to territory, this perspective means that managers and institutions may structure resources and territorial propositions, while actual value emerges through the practices, experiences, and uses of different audiences. For this reason, a territorial recommendation creates value not merely by identifying a probable alternative; it must also support conditions under which the territory becomes relevant to the objectives and audiences under consideration.

This understanding leads to the dimension of strategic coherence. A recommendation may be technically correct and functionally adequate, yet contribute little or even detract from territorial strategy. A system may, for example, repeatedly recommend the area with the highest probability of visitation even though local actors seek to reduce the concentration of flows and activate underused areas. In such a case, the recommendation accurately reproduces the observed tendency but reinforces a dynamic that contradicts the purpose for which the system was adopted.

Strategic coherence, therefore, refers to the degree to which the system's criteria, recommendations, and outputs align with the organizational, institutional, and territorial objectives that guide its use. Such objectives may include attraction, positioning, market selection, expansion, area prioritization, flow distribution, and territorial activation. They are defined and negotiated by networks of actors responsible for territorial management, not merely by the algorithm developer. Strategic coherence differs from functional adequacy: the former asks whether the recommendation advances the territorial purpose; the latter assesses whether the system can adequately perform the task required by that purpose.

Proposition 3. The quality of decision support provided by recommender systems in territorial marketing increases as their recommendations become more aligned with the attraction, prioritization, expansion, and activation objectives defined by the actors responsible for territorial management.

2.4 Frugal Recommender Systems in Territorial Marketing

The preceding discussions show that the evaluation of territorially situated recommender systems cannot be confined to accuracy alone. Recommendation research demonstrates that relevance depends on spatial, temporal, and contextual sensitivity. Territorial-marketing research shows that places are strategic, relational, and multifaceted entities. Frugal innovation and Green AI, in turn, question the disproportionate use of resources to obtain marginal performance gains. The contribution of this essay does not consist of fully integrating all these fields, but of relating specific concepts that address three inseparable questions: Does the system perform the required function? Are the resources consumed justifiable? Does the recommendation advance the territorial purpose?

Within MacInnis's (2011) typology, this contribution belongs to the broad category of relating and, more specifically, to its integrating subgoal. The conceptual move consists of connecting entities traditionally treated separately and revealing a shared latent structure. That structure is the evaluation of system value as a relationship among purpose, functioning, and material viability. The proposed integration problematizes the assumption that superior algorithmic performance, by itself, is equivalent to greater marketing value. Our structured literature mapping did not identify studies simultaneously integrating recommender systems, territorial marketing, and a frugal evaluative perspective, reinforcing the conceptual gap addressed by this essay.

2.4.1 Functional Adequacy

Functional adequacy derives from the intersection between spatially situated recommendation and frugal innovation. In POI research, recommendations become genuinely useful only when they incorporate the factors that constrain behavior in physical space. Yin *et al.* (2015) demonstrate the need to jointly model interests, semantic patterns, temporality, and mobility. Cai *et al.* (2018) show that spatial and temporal contexts improve performance and that the spatial dimension exerts a particularly relevant influence in the data analyzed. Baral and Li (2017) show that frequency, social relationships, time, space, and category play distinct roles and that combining them is more informative than considering any single aspect in isolation.

Frugal innovation complements this literature by shifting evaluation away from the continuous addition of capabilities and toward the preservation of core functionalities. Weyrauch and Herstatt (2016) establish three simultaneous criteria for characterizing

frugal innovation: substantial cost reduction, concentration on core functionalities, and an optimized level of performance. Optimized performance does not mean low performance, but rather a level appropriate to purpose and context, avoiding both insufficiency and overengineering. The concrete manifestation of these criteria depends on user needs and use conditions.

Applied to territorial recommender systems, functional adequacy is not synonymous with generic technical functioning. It requires correspondence between system configuration and the decision problem. A system may display high average accuracy yet remain inadequate if it ignores spatial constraints, territorial scale, temporality, accessibility, or the type of decision it is intended to support. Conversely, a less complex model may be functionally superior when it preserves essential components and produces recommendations that are sufficient, intelligible, and actionable in context.

The theoretical problem addressed by this dimension is algorithmic abstraction: the tendency to evaluate a system as though recommendation could be separated from space, time, and purpose of use. Functional adequacy reconnects recommendation to the physical world and the decision situation. It answers the question of how strategic purpose is translated into concrete requirements for modeling, data, and output.

Table 1: Theoretical Derivation of Functional Adequacy

Element	Synthesis
Foundational literatures	POI/LBSN recommendation and frugal innovation.
Key authors	Yin et al. (2015); Cai et al. (2018); Baral and Li (2017); Weyrauch and Herstatt (2016); Bhatti and Ventresca (2013).
Mobilized concepts	Spatial, temporal, and contextual sensitivity; mobility patterns; core functionalities; optimized performance; contextual sufficiency.
Addressed limitation	Abstract algorithm evaluation detached from the physical and decisional conditions of the territory.
Proposed integration	Functional quality depends on the correspondence between the system's essential capabilities and the concrete configuration of the territorial problem.
Definition	The degree to which the system's functionalities and outputs respond sufficiently and contextually appropriately to the territorial problem for which it was adopted.
Evaluative implication	Assess not only whether the system predicts accurately, but also whether it incorporates the necessary territorial conditions and produces outputs that are usable for decision-making.
Relationship with the other dimensions	Operationalizes the purpose defined by strategic coherence and must operate within the limits established by resource efficiency.

Source: Author's own elaboration based on the theoretical derivation developed in the essay.

2.4.2 Resource Efficiency

Resource efficiency derives from the convergence of Green AI, frugal innovation, and Frugal AI. Schwartz *et al.* (2020) use the term Red AI to describe an orientation that seeks performance improvements through the intensive expansion of computational power, data, and experimentation without assigning equivalent weight to their associated costs. The authors do not deny the scientific value of high-performing models; rather, their critique concerns the absence of efficiency as an evaluative criterion. Green AI therefore proposes making cost and efficiency visible alongside accuracy.

This critique is particularly relevant in the presence of diminishing returns, as linear or marginal performance gains may require disproportionately larger increases in complexity and processing. In this sense, performance and efficiency are not opposing concepts. The former describes the quality of the output, whereas the latter describes the cost required to obtain it. The theoretical issue is therefore not to choose between accuracy and efficiency, but to evaluate jointly the level of performance achieved and the resources mobilized.

In frugal innovation, substantial cost reduction must occur simultaneously with the preservation of core functionalities and an optimized level of performance (Weyrauch; Herstatt, 2016). Albert (2022), however, demonstrates that frugality cannot be automatically equated with environmental sustainability. The reviewed evidence is more consistent with economic and social effects, whereas ecological effects are more limited and context-dependent. This qualification prevents any cost reduction from being presented as an automatic environmental benefit.

Recent developments in Frugal AI broaden the unit of analysis from the individual model to the organizational lifecycle and portfolio. Gandikota *et al.* (2025) propose a technical ecosystem based on compression, efficient architectures, distributed orchestration, and open standards, reporting substantial reductions in total cost of ownership and energy use without a meaningful loss of model utility. Ahluwalia *et al.* (2025) structure evaluation across three levels: costs and efficiency at the TCO level, return on investment, and social impacts aligned with the Sustainable Development Goals.

Under this approach, resources include computational capacity, data, infrastructure, energy, emissions, development, integration, personnel, maintenance, monitoring, governance, and decommissioning. TCO encompasses initial, operational, and maintenance costs throughout the lifecycle. Metrics such as tokens per joule, cost per

token, carbon intensity, and normalized efficiency enable the distinction among financial, energy, and environmental performance, thereby preventing a single indicator from serving as an imperfect proxy for the others (Ahluwalia *et al.*, 2025).

Efficiency should therefore be understood as a relationship between the value produced and the resources consumed. Ahluwalia *et al.* (2025) define return on investment as the relationship between a value score and TCO and recommend that efficiency metrics inform decisions concerning optimization, scaling, and discontinuation. This reinforces the argument that resource savings do not possess value in themselves; they must be assessed in relation to utility, outcomes, and the organization's strategic objectives.

Resource efficiency thus refers to the contextually justifiable relationship between the value produced by the recommendation and the computational, energy, environmental, financial, and operational resources employed in its development, deployment, and use. It is not absolute. The appropriate level depends on the application's scale, risk, criticality, and purpose. A level of performance that is acceptable for an exploratory recommendation may be insufficient for a high-consequence decision; conversely, the pursuit of maximum precision may be unjustifiable when it does not alter the decision.

The theoretical problem addressed by this dimension is infrastructural waste resulting from an evaluative logic that celebrates outputs without examining their cost of production. Resource efficiency introduces proportionality; additional gains strengthen the solution only when the additional resources required are compatible with the incremental value obtained and with the material constraints of the context.

Proposition 2. While holding performance at a functionally sufficient level, a disproportionate increase in the consumption of computational, energy, financial, and operational resources reduces the viability of implementation and the frugal value of recommender systems applied to territorial decisions.

Table 2: Theoretical Derivation of Resource Efficiency

Element	Synthesis
Foundational literatures	Green AI, frugal innovation, and Frugal AI.
Key authors	Schwartz <i>et al.</i> (2020); Strubell <i>et al.</i> (2019); Weyrauch and Herstatt (2016); Albert (2022); Gandikota <i>et al.</i> (2025); Ahluwalia <i>et al.</i> (2025).
Mobilized concepts	Efficiency as an evaluative criterion; diminishing returns; substantial cost reduction; TCO; ROI; energy efficiency; lifecycle; optimized performance.

Element	Synthesis
Resources considered	Computing, data, infrastructure, energy, emissions, financial costs, personnel, integration, maintenance, monitoring, governance, and decommissioning.
Addressed limitation	Performance-oriented evaluation that renders invisible the material and operational costs required to produce the outcome.
Proposed integration	The quality of the solution depends on the relationship between the value produced and the resources consumed, rather than on the isolated minimization of costs or the isolated maximization of accuracy.
Definition	The contextually justifiable relationship between the value of the recommendation and the resources mobilized throughout the system lifecycle.
Evaluative implication	Compare performance, utility, TCO, energy consumption, emissions, and operational costs, and identify the point at which marginal gains no longer justify additional resource use.
Relationship with the other dimensions	Constrains the functional and strategic operationalization of the system according to the material limits and viability of the ecosystem.

Source: Author's own elaboration based on the theoretical derivation developed in the essay.

2.4.3 Strategic Coherence

Strategic coherence derives primarily from the territorial marketing literature. Kavaratzis (2005) shows that place marketing evolved toward an integrated management orientation in which attractiveness, infrastructure, services, communication, and positioning must form a coherent proposition. Warnaby (2009) demonstrates that place is a complex product and that its management depends on equally complex organizational mechanisms. Warnaby and Medway (2013) add that territory combines materiality with changing narratives produced by actors holding distinct interpretations and interests.

These contributions support the argument that places cannot be evaluated solely as points characterized by measurable attributes. They are strategic objects composed of resources, actors, structures, uses, experiences, and meanings. Service-Dominant Logic, used here as a complementary perspective, helps explain that managers structure propositions and conditions of use, while value is realized through audience experience and resource integration. Grönroos's (2011) contribution is especially important because it clarifies that co-creation does not occur automatically; rather, it depends on interactions that allow the provider to participate in the user's value-creation process.

For recommender systems, the implication is that an accurate prediction does not necessarily guarantee marketing value. A system may identify the alternative with the highest likelihood of being chosen and still conflict with objectives related to development, flow distribution, audience attraction, or area activation. Strategic coherence evaluates whether technology functions as a means to a territorial end, rather than allowing predictive maximization to become an autonomous objective.

The theoretical problem this dimension addresses is the marketing blindness inherent in instrumental evaluation. Strategic coherence connects algorithmic outputs to the decisions that justify the system's existence. It establishes why the system exists, defines what should count as success, and provides the criterion for interpreting whether the recommendation creates value for both the organization and the territory.

Table 3: Theoretical Derivation of Strategic Coherence

Element	Synthesis
Foundational literatures	Territorial marketing, place marketing, and the relational perspective on value.
Key authors	Kavaratzis (2005); Warnaby (2009); Warnaby and Medway (2013); Vargo and Lusch (2004, 2008); Grönroos (2011).
Mobilized concepts	Strategic place management; attractiveness; positioning; multiplicity of actors; resources; experience; value-in-use; territorial activation.
Addressed limitation	Instrumental evaluation that treats predictive accuracy as an end in itself without examining its contribution to territorial objectives.
Proposed integration	Recommendation generates marketing value when its outputs and criteria support the objectives that motivated system adoption.
Definition	The degree of alignment between the system's criteria, recommendations, and outcomes and the organizational, institutional, and territorial objectives underlying its adoption.
Evaluative implication	Assess who defines the objectives, which territorial effects are intended, and whether the recommendations contribute to attraction, prioritization, expansion, flow distribution, and activation.
Relationship with the other dimensions	Establishes the purpose; functional adequacy operationalizes it; resource efficiency conditions its realization.

Source: Author's own elaboration based on the theoretical derivation developed in the essay.

2.4.4 Articulation Among the Dimensions

The three dimensions do not constitute a list of independent attributes. Rather, they form a logical and interdependent structure. Strategic coherence establishes the system's purpose by defining which territorial problem should be supported and which outcomes constitute value. Functional adequacy translates this purpose into requirements concerning data, modeling, and outputs that are sensitive to space, time, and context. Resource efficiency, in turn, conditions the solution according to the computational, energy, financial, and operational constraints of the ecosystem.

This ordering constitutes an integrative proposition for this essay rather than a causal sequence previously formulated within a single body of literature. Its purpose is to make visible the latent structure produced by the conceptual integration: purpose, operationalization, and material constraint. The absence of any one dimension compromises the others. Without strategic coherence, the system may optimize an irrelevant objective; without functional adequacy, the intended purpose cannot be

translated into a viable recommendation; and without resource efficiency, the solution may be technically adequate and strategically desirable yet unfeasible to develop, scale, or maintain.

The framework's parsimony derives from this coverage. The three dimensions correspond to the fundamental aspects of technology adoption in territorial problems: strategic purpose, contextual functioning, and material viability. Additional aspects may be treated as indicators or internal subdimensions, but elevating them to the same conceptual level would generate overlap. The proposal does not assume that all indicators are equivalent; rather, it argues that a frugal evaluation must address all three questions simultaneously.

2.5 Conceptual Model and Integrative Proposition

The conceptual model represents frugal evaluation as a relational structure rather than as a list of desirable attributes. Its point of departure is the problematization of the presumed equivalence between algorithmic performance and marketing value. The model proposes that the frugal value of a territorial recommender system depends on the articulation among strategic coherence, functional adequacy, and resource efficiency, each of which is associated with a specific consequence for territorial decision support.

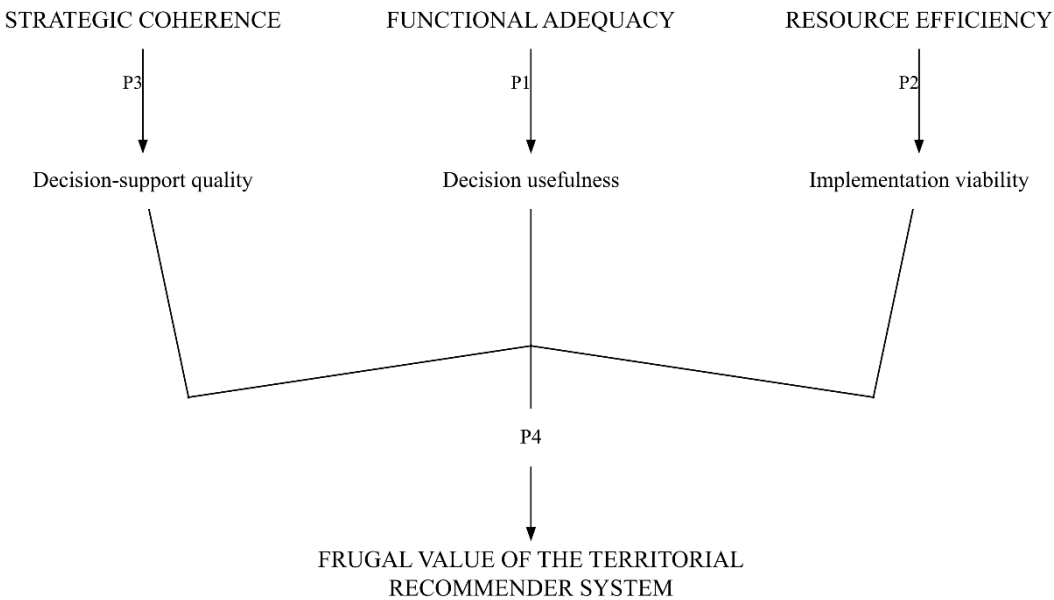
Strategic coherence has logical precedence because it defines the territorial problem, the relevant actors, and the outcomes that constitute value. It is associated with decision-support quality because alignment with territorial objectives determines whether the recommendation contributes to attraction, prioritization, expansion, flow distribution, or activation. Functional adequacy operationalizes this purpose by translating it into requirements concerning data, modeling, and outputs that are sensitive to spatial, temporal, contextual, and decisional conditions. It is associated with decision usefulness because a technically correct output has limited value if it does not address the concrete problem. Resource efficiency operates across the model by assessing whether the value produced justifies the computational, energy, financial, and operational resources mobilized. It is associated with implementation viability because disproportionate requirements may hinder the development, scaling, maintenance, or continued use of the system.

The model does not assume an antagonism between performance and efficiency. Accuracy, ranking quality, robustness, and other technical metrics remain necessary, but they are interpreted in relation to the decision problem, the cost of obtaining the result,

and the strategic purpose. The relevant question is no longer which model achieves the highest isolated performance, but which solution provides sufficient and territorially meaningful decision support using justifiable resources.

Proposition 4. The frugal value of recommender systems applied to territorial marketing results from the joint and interdependent presence of functional adequacy, resource efficiency, and strategic coherence; deficiencies in any one of these dimensions reduce the overall quality of the solution and constrain its contribution to territorial decision-making.

Figure 1: Relational model of the frugal value of territorial recommender systems



Source: Author's own elaboration (2026).

Figure 1 represents the relational structure of the proposed framework. Strategic coherence is associated with decision-support quality because it captures the alignment between system recommendations and the territorial objectives that motivate adoption. Functional adequacy is associated with decision usefulness because it reflects the fit between system capabilities and the spatial, temporal, contextual, and decisional conditions of the problem. Resource efficiency is associated with implementation viability because it evaluates whether the value produced justifies the computational, energy, financial, and operational resources mobilized. The joint effects of these three dimensions expressed through decision-support quality, decision usefulness, and implementation viability constitute the frugal value of the territorial recommender

system. Thus, the model does not treat the dimensions as independent evaluation criteria, but as complementary and interdependent conditions whose specific and combined effects shape the overall quality of the solution.

In terms of contribution, the essay develops a relating contribution, specifically within the integrating subgoal identified by MacInnis (2011). It integrates recommender systems with territorial marketing through a frugal evaluative perspective informed by Green AI, Frugal AI, and frugal innovation. By bringing these theoretical streams into dialogue, the essay reveals an evaluative structure that remains obscured when they are considered separately and demonstrates that algorithmic performance becomes marketing value only when it is functionally adequate, materially justifiable, and strategically coherent.

3. CONCLUSION

The growing incorporation of artificial intelligence into marketing has consolidated an analytical rationality oriented toward automation, personalization, and performance at scale. In recommender systems, this orientation is reflected in the centrality of accuracy, ranking quality, predictive capability, robustness, and scalability. These criteria are necessary, but insufficient to define the value of systems that support decisions concerning places, regions, and markets.

This essay problematizes the assumption that superior algorithmic performance automatically generates greater marketing value. In territorial contexts, recommendation is conditioned by space, time, accessibility, mobility, and decision purpose. Producing the output also requires computational, energy, financial, and operational resources. Moreover, places are not neutral entities; they are strategic, multifaceted entities composed of resources, actors, structures, uses, experiences, and meanings.

The central contribution is to propose frugality as a theoretical lens for reorganizing this evaluation. Frugality does not mean accepting inferior technology or replacing accuracy with cost reduction. Rather, it introduces a logic of proportionality and sufficiency: the system should preserve core functionalities, achieve a level of performance appropriate to the context, and justify the resources used in relation to the value produced. Green AI makes computational and energy costs visible; frugal innovation focuses on essential functionalities and optimized performance; and Frugal AI extends the analysis to the lifecycle, total cost of ownership, organizational return, and portfolio-level impact.

The framework derives three interdependent dimensions. Strategic coherence defines why the system is adopted and which territorial effects it should support. Functional adequacy determines how these objectives are translated into recommendation capabilities that are sensitive to the concrete conditions of the territory. Resource efficiency constrains this operationalization according to material limits and the viability of development, deployment, scaling, and maintenance. The articulation among the three dimensions reveals a latent structure organized around purpose, functioning, and material viability.

The distinction between consumer-oriented POI recommendation and organizational territorial decision-making is essential to delimiting the argument. Systems designed to help individuals discover places provide important foundations concerning mobility, context, and spatiality, but they are not automatically territorial marketing systems. The framework focuses on systems that support organizations and territorial actors in comparing, selecting, prioritizing, expanding, and activating territorially situated alternatives.

The theoretical contribution lies not only in asserting that three dimensions are relevant, but also in specifying relationships between constructs and consequences. Decision usefulness increases with functional adequacy; disproportionate resource consumption reduces implementation viability and frugal value; decision-support quality increases with strategic alignment; and the overall value of the solution depends on the joint presence of the three conditions. The essay thus moves beyond descriptive classification toward a relational and normative conceptual formulation.

The proposal also preserves important boundaries. Frugality is not automatically synonymous with environmental sustainability, whose effects remain context-dependent. Service-Dominant Logic is used as a complementary perspective rather than as a consensual position within territorial marketing. The proposed relationships still need to be operationalized and empirically examined, especially in organizational systems used to compare and prioritize municipalities, regions, and markets.

Future research may develop measures for each construct, compare exhaustive and frugal models, identify thresholds of sufficient performance, and examine how territorial actors negotiate the objectives embedded in recommender systems. It may also investigate tensions among the dimensions, such as situations in which an efficient and functionally accurate recommendation reinforces flows that run counter to territorial

strategy, or in which a strategically desirable solution requires resources that are incompatible with its scale of use.

In summary, better territorial recommendations are not necessarily those produced by the most complex model or by the greatest amount of processing. They are those that respond appropriately to the problem, use resources in proportion to the value produced, and remain aligned with the objectives that justified their adoption. This combination transforms algorithmic performance into frugal value for territorial marketing.

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